Tema 4 PMP

Ex 1. a) Conditia $A \perp C | B$ înseamnă că variabilele aleatoare A și C sunt independente conditionat de B, adică, dacă știm valoarea lui B, cunoașterea valorii lui C nu ne oferă informații suplimentare despre A.

b) $P_{(a,b,c)} = P_{(a|b,c)} \cdot P_{(c|b)} \cdot P_{(b)}$ |

$A \perp C | B \Rightarrow P_{(a|b,c)} = P_{(a|b)}$ | $\Rightarrow$

$\Rightarrow P_{(a,b,c)} = P_{(a|b)} \cdot P_{(c|b)} \cdot P_{(b)}$ |

$\phi_{AB}(a,b) = P_{(a|b)}$

$\phi_{BC}(b,c) = P_{(c|b)} \cdot P_{(b)}$ | $\Rightarrow$

$\Rightarrow P_{(a,b,c)} = \phi_{AB}(a,b) \cdot \phi_{BC}(b,c)$

$P_{(a,b,c)} = \frac{1}{Z} \phi_{AB}(a,b) \phi_{BC}(b,c)$

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c) $P_{(a,b,c)} = \frac{1}{Z} \phi_{AB}(a,b)\, \phi_{BC}(b,c)$

$$P_{(a,c|b)} = \frac{P_{(a,b,c)}}{P_{(b)}}$$

$$P_{(a,c|b)} = \frac{\frac{1}{Z}\phi_{AB}(a,b)\phi_{BC}(b,c)}{P_{(b)}}$$

$$P_{(b)} = \frac{1}{Z}\left(\sum_{a'} \phi_{AB}(a',b)\right)\left(\sum_{c'} \phi_{BC}(b,c')\right)$$

$$P_{(a,c|b)} = \frac{\phi_{AB}(a,b)}{\sum_{a'} \phi_{AB}(a',b)} \cdot \frac{\phi_{BC}(b,c)}{\sum_{c'} \phi_{BC}(b,c')}$$

$$P_{(a,c|b)} = P_{(a|b)} \cdot P_{(c|b)} \Rightarrow A \perp C \mid B$$

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